

3D ITS Campus on the Web: A WebGL Implementation

Anny Yuniarti, Ardian Atminanto, Anggi Mardasatria, Ridho Rahman Hariadi, Nanik Suciati

The Interaction, Graphics and Art Laboratory, Department of Informatics

Institut Teknologi Sepuluh Nopember (ITS)

Surabaya, Indonesia

anny@if.its.ac.id

Abstract—Multimedia web experience can be increased using 3D graphics on the web. Proprietary systems need specific plug-ins to download and install. Therefore, open standard methods have been used increasingly. WebGL is one of those that is a cross-platform, web standard for a 3D graphics API based on OpenGL ES 2.0. There are many other Javascript libraries that abstract WebGL and results in higher code level. One of the libraries is Three.js, the most popular library/API for WebGL-based 3D graphics. This paper proposed an implementation of WebGL and Three.js libraries for a 3D ITS Campus information system. The proposed system can be viewed using two modes: ground and free modes. In the ground view mode, users can not move in vertical direction. Whereas the free view mode enables users to move in any direction. A usability testing was performed after development. The participants consist of 12 postgraduate students who are currently studying at the Informatics Department ITS, graduated from both ITS and non-ITS campus in their previous study. Results from the experiment show that the proposed system effectively render the ITS campus (83% of participants successfully view the 3D ITS campus on their own computers) and most participants suggest to add more buildings and natural objects like trees and ponds to the system.

Keywords—WebGL; 3D graphics; 3D information system; Three.js; ITS Campus.

I. INTRODUCTION

3D graphics has become more important in enhancing multimedia web experience. It was begin with the creation of X3D standard using a declarative approach in order to create 3D graphics on the web. Now, imperative approaches have enabled developers to create web-based 3D graphics applications more flexibly.

Initially, interactive multimedia content can be produced using Adobe Flash. In order to access audio, video, and interactive 2D animation on the web, users have to download and install a Flash supporting browser plugin first. Flash based applications generally need a lot of download time and it is not available in iOS platform [1]. Consequently, open standard methods for 3D graphics on the web have been increasingly used.

The campus of ITS (Institut Teknologi Sepuluh Nopember) is located in Surabaya, Indonesia. The ITS campus has very wide area, i.e around 18,000 m², contains five faculties and 26 schools. Therefore, future students of ITS, visiting researchers or guests of ITS need more information about locations in ITS campus before they physically go to the ITS campus. Thus, a web-based 3D graphics of ITS campus can increase interests of future students or guests of ITS.

This paper proposed a 3D ITS Campus information system based on WebGL. A usability testing has been conducted in order to gather information from testing participants about the proposed system. The rest of this paper is organized as follows. The literature review of browser-based 3D rendering and 3D web applications is presented in Section 2. The proposed system implementation, evaluation and usability testing are described in Section 3. Finally, some conclusions and future works are suggested.

II. LITERATURE REVIEW

A. Browser-based 3D Rendering

There are two groups in browser-based 3D rendering methods: declarative and imperative [2]. Each group can be further divided into 2D and 3D cases. An example of a declarative approach for 2D is Scalable Vector Graphics (SVG). Whereas for the imperative approach we can use Javascript and <canvas> element. However, this categorization cannot be strictly used in 3D cases since many imperative techniques use a lot of Scene Graph that can be considered as a declarative approach [1].

B. 3D Web Applications

3D web applications have some differences with usual (hypertext only) web applications. Tasks performed in 3D web applications are richer than usual ones, for example, not only users can click on hyperlinks or find texts on a page, but also users can navigate 3D scenes or select objects [1].

Major uses of 3D web applications are for data visualization and natural/architecture/virtual heritage. A 3D web application that visualize 3D CAD data using X3D standard was developed [3]. In the proposed system, the 3D

data that include geometry, structures, and manufacturing are extracted from a CAD system and translated into X3D standard. The proposed system visualized better than that using VRML standard [3].

An application for remotely visualizing astronomy images was developed using an existing open source project, namely, IIPImage¹, a client-server system for web-based streamed viewing and zooming of ultra high-resolution raster images [4]. This application follows the remote rendering concept, i.e. involving server-based data generation which is then passed to a client for visualisation.

In a geography application field, there is OWGIS [5], an open source Java web application to display data from map servers. The system uses Java 7 and OpenLayers 2 to build maps. Another application in geography field is the mobile 3D-GIS hybrid recommender system for tourism by Noguera et. al [6]. In the proposed system, the client application used a C++ pro-gram and OpenGL ES. Moreover, Hsu investigated the use of Second Life (SL), a 3D web graphics application, for tourism education [7].

Li and Zhu [8] proposed a web-based information system for shield tunnel construction projects. The proposed system, developed based on ArcGIS Server, is for managing, visualizing, and analyzing shield tunnel construction data.

Finally, a campus information system based on GLGE has been introduced by Hering et.al [9]. However, the proposed system needs additional software, i.e Blender-to-GLGE exporter, that is not easily available. In addition, the system was still not finished and is still in further development [9].

III. SYSTEM IMPLEMENTATION AND EVALUATION

A. System Prototype

At the beginning of our development stage, we created a prototype that only shows one ITS building, namely, the Informatics department building. The 3D model was created using Blender and the 3D data was exported in Collada (Collaborative Design Activity) standard (.dae file extension). Collada has been used widely, namely by Google Earth, web 3D engines such as Papervision3D, Google O3D, and Google 3Dwarehouse.

However, when the exported 3D data rendered using WebGL, the quality of the viewed model was significantly reduced compared to the original view in Blender. Fig. 1 shows the Informatics building viewed in Blender, whereas Fig. 2 shows the rendered building viewed in web. Images in Fig. 1 has higher level of detail than that in Fig. 2. For the implementation, we used another 3D format, namely .obj and .mtl file formats.



Fig. 1. The Informatics building viewed in Blender.



Fig. 2. The same Informatics building viewed in web.

B. Implementation and Evaluation

The proposed system was implemented using Three.js library. First, a scene was created:

```
var scene = new THREE.Scene();
```

Next, a virtual camera was added at the position of the ITS Campus main entrance (-3.31048, 0.05, -1.74981). The camera used a perspective view with 45° field of view (fov), and the distances between the camera and the front and back image planes are 0.1 and 100 respectively:

```
var camera = new THREE.PerspectiveCamera(45,
    window.innerWidth / window.innerHeight, 0.1,
    100);
camera.position.set(-3.31048, 0.05, -1.74981);
```

This camera is used by FirstPersonControls to control the camera movement. We set the mouse speed to look around the scene to 0.075, and the movement speed of the camera to 1.0:

```
var camControls = new
    THREE.FirstPersonControls(camera);
camControls.lookSpeed = 0.075;
camControls.movementSpeed = 1.0;
```

There are two different modes for viewing: ground view and free view. In the ground mode, the camera's y position is fixed. Thus, viewers cannot move up or down (Fig. 3). Whereas in the free mode, viewers are able to look everywhere (Fig. 4).

¹ <http://iipimage.sourceforge.net>.

The landscape was created using texture mapping. First, a grass image was loaded. Then, we repeat the texture horizontally and vertically. Next, a plane geometry and the mesh was created and positioned below $y = 0$, and finally added to the scene.

The display of the proposed system is divided into three regions: left, top right and bottom right (see the sample layout on Fig 5). The left region is for displaying information and provides links of ITS buildings. The top right is for displaying the 3D ITS Campus and the bottom right displays instructions for users. The proposed system provides interaction to obtain building information by clicking a symbol above each building or a link at the left menu. All information is stored at JSON files. As a trigger to display certain information, we created an information symbol placed above each ITS building (Fig. 6). Sample information display is as shown in Fig 7.

The 3D ITS building models was designed using Blender version 2.72 and consisted of 44 buildings in ITS Campus. The models are then exported into obj and mtl files. In order to measure the rendering performance, we measured the object load time based on the object file size (Fig. 8).



Fig. 3. Example of a ground view of ITS Campus.



Fig. 4. Example of a free view of ITS Campus.

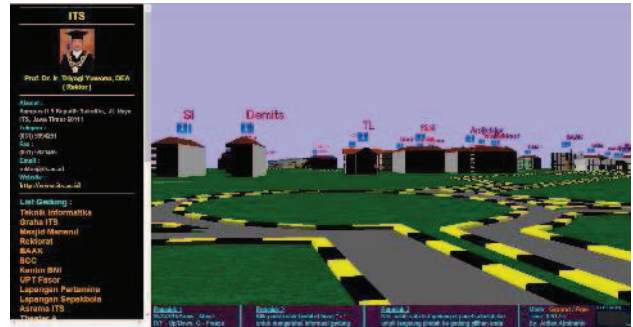


Fig. 5. The layout of the proposed system.



Fig. 6. The information symbol placed above each ITS building.



Fig. 7. Sample display of Informatics Department information.

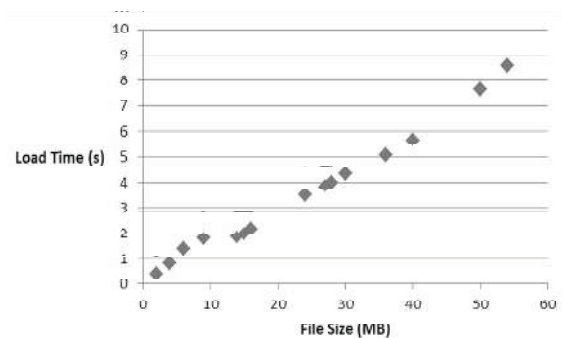


Fig. 8. Object load time (s) based on its file size (MB).

We also compare the results using several browsers: (1) Google Chrome version 39, (2) Mozilla Firefox version 34.0, (3) Opera version 26.0, (4) Internet Explorer version 10, and (5) Internet Explorer version 11. The results show that almost all browsers are able to render correctly except the Internet Explorer version 10.

C. Usability Testing

In order to gather responds from real users, we performed a usability testing after the development. The testing participants are postgraduate students of Informatics Department ITS consisting of 7 males and 5 females. The age ranges from 22 to 30 years old. Six students graduated from the Informatics Department ITS. The others graduated from institutions other than Informatics ITS at Surabaya (East Java), Gresik (East Java), Makasar (South Sulawesi), and Tomohon (North Sulawesi). This represents the variation of user background.

In addition, we asked whether participants ever use systems that include 3D world navigation tasks. From the background questionnaire only four participants have ever used 3D systems, the other eight participants have never tried 3D systems before.

In the usability testing, we asked participants to visit the proposed system at the given URL. Then we asked several questions, such as whether the 3D models rendered successfully and the browser types used by participants. There are two participants who could not view the system at their computers. Both used Mozilla Firefox browser, but unfortunately they did not provide the version number at the questionnaire form. This means 83% of the participants successfully view the 3D ITS Campus on the web.

The browsers successfully used by participants to view the 3D ITS Campus on the web varied: Mozilla Firefox (8 users), Google Chrome (3 users), and Opera (1 user). No participants used the Internet Explorer to successfully view the 3D model on the web. This was because participants used the version number of IE that still does not support WebGL.

IV. CONCLUSIONS

A WebGL-based 3D ITS campus information system was successfully developed using Blender and the Three.js library. The initial prototype system using Collada (.dae format) standard resulted in a lot of difference between views in Blender and views in Web. Therefore we conclude that the .obj and .mtl 3D file formats suit better for the proposed system. The usability testing shows that users found the system to be interesting but need improvements especially that it needs additional buildings and surroundings such as trees and ponds.

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